

Gym Membership Login System

Software Requirements Specification

CS 4398

Software Engineering Project

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1. Introduction

1.1 Document Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive, detailed description of the Gym Management System (GMS). This document defines both the functional and non-functional requirements of the system to ensure a clear understanding between the development team and the project stakeholders. It outlines the system's architecture, user interfaces, database interactions, and operational constraints, serving as the primary blueprint for the design, implementation, and testing phases of the project.

1.2 Product Scope

The Gym Management System is a centralized software solution designed to streamline the daily operations of a modern fitness facility. The software replaces manual, paper-based tracking with an automated digital ecosystem to manage memberships, attendance, staff scheduling, and financial transactions.

- **In-Scope Capabilities:**
 - **User Management & Authentication:** Secure registration and multi-tier login credentials for administrators, trainers, and gym members.
 - **Membership Management:** Creation, modification, renewal, and suspension of membership subscription tiers (e.g., monthly, annual, VIP).
 - **Attendance Tracking:** Automated member check-in and check-out logs using unique digital identifiers or scanning codes.
 - **Payment Processing:** Integration of secure payment processing to handle membership billing, recurring payments, and manual point-of-sale invoices.
- **Out-of-Scope Capabilities:**
 - Direct physical hardware control (e.g., physical turnstile gate mechanics, though the system will trigger software approval signals).
 - Full-scale accounting and payroll (the system logs transactions but does not replace professional accounting software like QuickBooks).

1.3 Intended Audience

This document is structured for several key audiences involved in the software lifecycle:

- **Project Stakeholders / Gym Owners:** To review and verify that the system scope meets their operational needs and business goals.
- **Software Development Team:** To guide the design, database modeling, API development, and frontend implementation of the gym platform.
- **Quality Assurance (QA) & Testing Team:** To formulate test plans, test cases, and validation criteria that ensure the system functions strictly as specified.
- **System Administrators & Staff:** To understand the administrative capabilities and boundaries of the application they will eventually operate.

1.4 Definitions, Acronyms, Abbreviations

Term	Definition
Member	A registered customer of the fitness center.
Receptionist	Staff member responsible for member registration and daily operations.
Administrator	User with complete system management privileges.
Membership	Subscription that grants access to gym facilities.
Attendance	Record of member check-in and check-out.
Booking	Reservation for a fitness class.
Payment	Transaction made for membership purchase or renewal.
RFID	Radio Frequency Identification used for member check-in.
SRS	Software Requirements Specification.

1.5 Overview / Document Conventions

This SRS is organized into four core sections based on IEEE software documentation standards:

1. **Section 1: Introduction** defines the purpose, scope, target audience, and standard vocabulary of the system.
2. **Section 2: Overall Description** (to be completed next) will cover the high-level system perspective, operating environment, and system constraints.

3. **Section 3: Functional Requirements** outlines detailed system behavior, using UML diagrams (Use Case, Sequence, Class, ERD) and precise Use Case specification tables.
4. **Section 4: Non-Functional Requirements** details operational performance, security protocols, reliability expectations, and system usability constraints.

2. Overall Description

2.1 Document Perspective

The Gym Membership Login System is a standalone application developed to improve the management of fitness center operations. The system replaces manual record keeping with a centralized software solution that stores member information, attendance records, membership details, payments, and class schedules.

The software interacts with external services such as payment gateways and email notification services while maintaining a secure database of all operational information.

2.2 Document Functions

The Gym Management System provides administrators, staff, and gym members with a suite of automated operations. The core functions of the system include:

- **Member Registration:** Creating new member accounts, capturing profile details, and assigning unique membership identifiers.
- **Authentication & Access Control:** Processing user logins (Admin, Trainer, Member) and regulating feature access based on roles.
- **Membership & Subscription Management:** Assigning, updating, renewing, or suspending various membership plans and recording associated payments.
- **Attendance Tracking:** Logging member check-in and check-out timestamps dynamically to monitor daily facility capacity.
- **Trainer Scheduling:** Allowing trainers to manage their class times, and allowing members to view and enroll in available fitness sessions.

2.3 User Characteristics

The system accommodates three distinct user roles, each requiring varying levels of technical familiarity and system access:

- **Administrator (Admin):** High technical proficiency. Admins maintain complete control over the application. They manage user registration, modify system configurations, process payments, edit subscription rates, and generate operational audits.
- **Trainer (Staff):** Moderate technical proficiency. Trainers are gym staff members who use the system to manage daily workout schedules, update class capacities, and check member attendance.
- **Gym Member:** General/Low technical proficiency. Members use the front-end interface to check their personal subscription status, view workout/class schedules, and log their entry into the gym.

2.4 User Interface

The system features a clean, responsive Graphical User Interface (GUI) tailored to the specific user logged in:

- **Admin Dashboard:** A highly analytical interface highlighting total active members, daily attendance metrics, and pending renewals.
- **Trainer/Staff Portal:** A schedule-centric view focusing on calendar events, upcoming classes, and registered class participants.
- **Member View:** A simple, mobile-friendly interface showcasing active membership details, check-in history, and a digital ID code for scanning.

2.5 End-User Operating Environment

To ensure maximum accessibility, the Gym Management System is cross-platform and will operate reliably in the following environments:

- **Client-Side Operating Systems:** Windows 10/11, macOS (12.0 or higher), iOS, and Android mobile platforms.
- **Supported Browsers:** Modern, standard web browsers including Google Chrome (v100+), Mozilla Firefox (v98+), Apple Safari (v15+), and Microsoft Edge.
- **Database Environment:** A relational database management system (such as MySQL or PostgreSQL) to secure user data and transactional logs

2.6 Design and Implementation Constraints

The following constraints apply to the system:

- Users must authenticate before accessing protected features.
- The system shall use a MySQL relational database.
- Java shall be used as the primary programming language.
- JDBC shall provide database connectivity.
- Internet access is required for payment gateway communication.
- User passwords shall be encrypted before storage.

3. Functional Requirements

3.1 Summary

The Gym Membership Login System automates the management of members, memberships, attendance, payments, and fitness class reservations. The application reduces manual work performed by receptionists and administrators while improving accuracy, security, and efficiency.

Development tools include:

- Git for Version Control
- GitHub for Repository Management
- JUnit for Unit Testing
- MySQL Database
- Java
- JDBC
- StarUML / Visual Paradigm for UML Modeling

3.2 Use Case Diagram

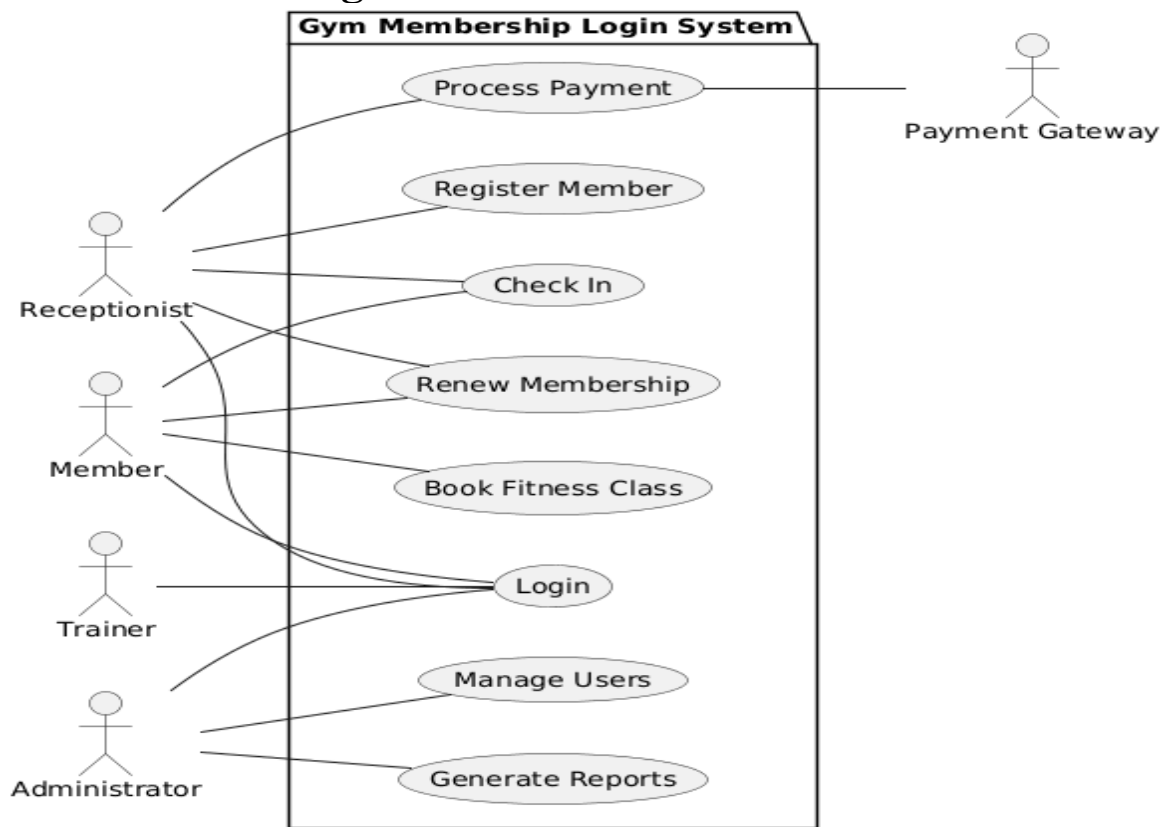


Figure 1: Use Case Diagram for the Gym Membership Login System

3.2.1 Use Case: Receptionist – Register Member

Name of the Use Case

Register Member

Brief Description

The receptionist registers a new member by entering personal information, selecting a membership plan, and creating a new account.

Preconditions

- Receptionist is logged in.
- Internet and database connection are available.

Postconditions

- Member account is created.
- Unique Member ID is generated.
- Membership becomes active.

Events Flow

1. Receptionist selects **Register Member**.
2. Enters member information.
3. Selects membership type.
4. Saves member record.
5. System generates Member ID.
6. Confirmation page displayed.

Alternate Flow and Exceptions

- Missing required information.
- Duplicate email address.
- Database unavailable.

3.2.2 Use Case: Member – Login

Name of the Use Case

Member Login

Brief Description

The member authenticates using a username and password to access the Gym Membership Login System.

Preconditions

- Member account exists.
- Internet connection available.

Postconditions

- Member dashboard displayed.

Events Flow

1. Enter username.
2. Enter password.
3. Click Login.
4. System validates credentials.
5. Dashboard displayed.

Alternate Flow and Exceptions

- Invalid username or password.
- Membership expired.
- Account suspended.

Absolutely! Below is the **second half** of your **Software Requirements Specification (SRS)** for the **Gym Membership Login System**, following the same organization and writing style as the Classified Site SRS example.

3.2.3 Use Case: Member – Check In

Name of the Use Case

Member Check In

Brief Description

This use case allows a member to check into the gym using their membership ID, barcode, or RFID card. The system verifies that the membership is active before recording attendance.

Preconditions

- Member has an active membership.
- Member is registered in the system.

Postconditions

- Attendance record is created.
- Check-in time is stored.
- Member is granted access to the gym.

Events Flow

1. Member presents membership card.
2. Receptionist scans barcode or RFID.
3. System retrieves member information.
4. Membership status is verified.
5. Attendance record is created.
6. Confirmation message is displayed.

Alternate Flow and Exceptions

- Membership has expired.
- Membership is suspended.
- Scanner unavailable.
- Database connection failure.

3.2.4 Use Case: Member – Book Fitness Class

Name of the Use Case

Book Fitness Class

Brief Description

Members may reserve available fitness classes through the system.

Preconditions

- Member is logged in.
- Membership is active.
- Class has available seats.

Postconditions

- Reservation is stored.
- Member receives confirmation.

Events Flow

1. Member selects **Fitness Classes**.
2. Available classes are displayed.
3. Member selects a class.
4. System verifies availability.
5. Reservation is saved.
6. Confirmation is displayed.

Alternate Flow and Exceptions

- Class is already full.
- Member has already booked the class.
- Membership has expired.

3.2.5 Use Case: Receptionist – Process Membership Payment

Name of the Use Case

Process Membership Payment

Brief Description

Receptionist processes payment for a new membership or membership renewal.

Preconditions

- Member account exists.
- Membership plan selected.

Postconditions

- Payment recorded.
- Membership activated or renewed.
- Receipt generated.

Events Flow

1. Receptionist selects **Process Payment**.
2. Membership information is displayed.
3. Payment amount calculated.
4. Member selects payment method.
5. Payment submitted.
6. Payment gateway authorizes transaction.
7. Database updated.
8. Receipt printed.

Alternate Flow and Exceptions

- Payment declined.
- Payment gateway unavailable.
- Network failure.

3.3 Sequence Diagram

The following sequence diagrams describe interactions between users and the Gym Membership Login System.

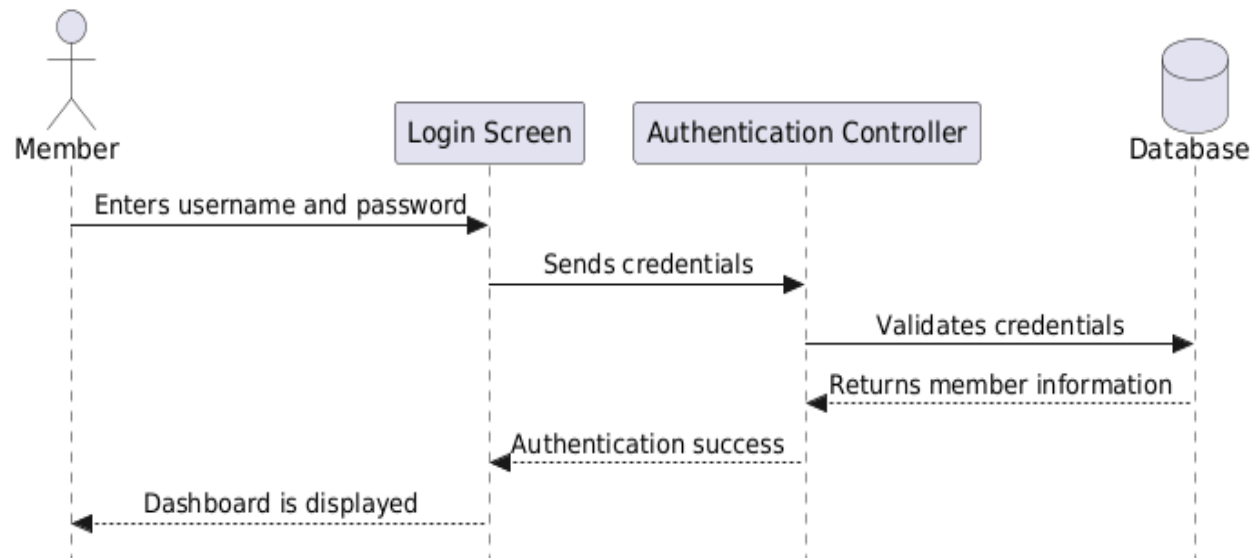


Figure 2: Member Login Sequence

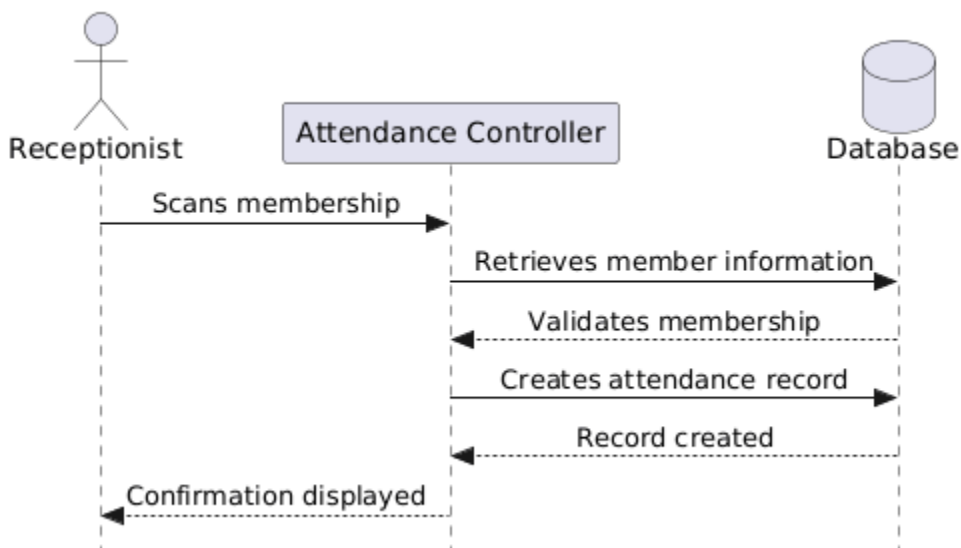


Figure 3: Member Check-In Sequence

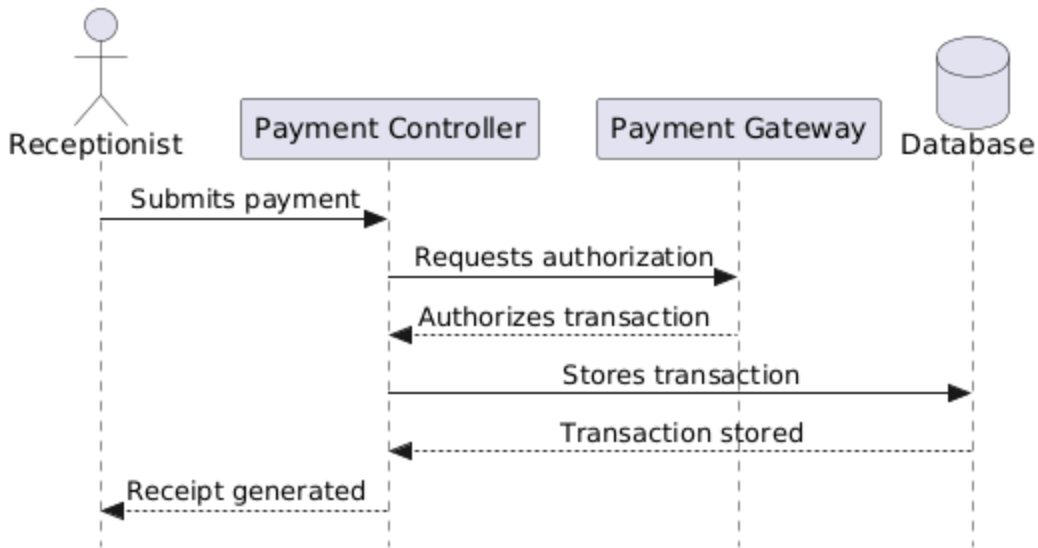


Figure 4: Payment Processing Sequence

Actors

- Receptionist
- Payment Controller
- Payment Gateway
- Database

Sequence

1. Receptionist submits payment.
2. Payment Gateway authorizes transaction.
3. Database stores transaction.
4. Receipt generated.

3.4 Entity Relationship Diagram

The Gym Membership Login System stores information using a relational database.

Main Entities

- Member
- Membership
- Attendance
- Booking

- Payment
- Trainer
- FitnessClass
- UserAccount

Relationships

- One Member has one Membership.
- One Member has many Attendance records.
- One Member has many Bookings.
- One Trainer teaches many Fitness Classes.
- One Membership has many Payments.

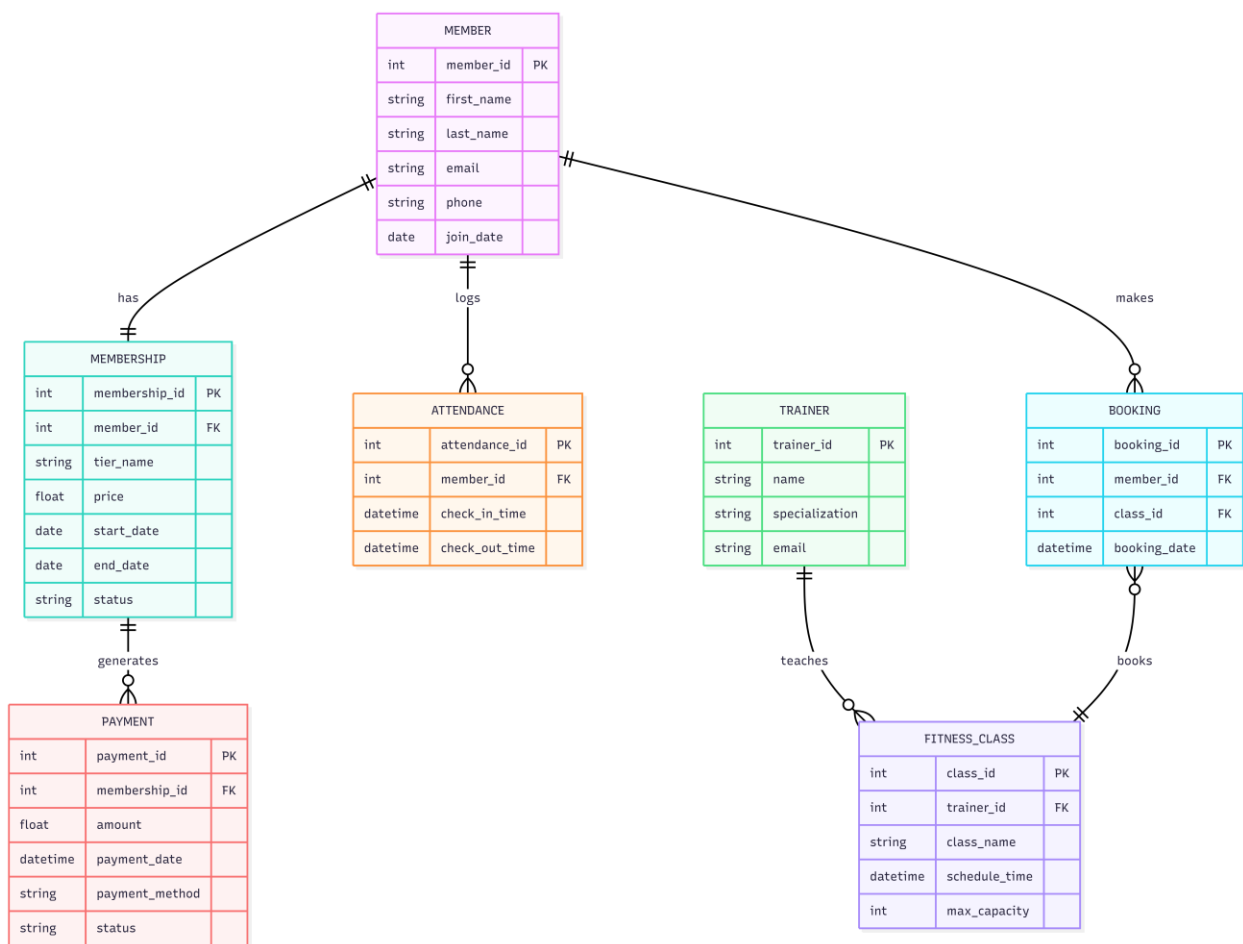


Figure 5: Entity Relationship Diagram

3.5 Class Diagram

The Class Diagram illustrates the static structure of the Gym Membership Login System.

Main Classes

- Member
- Membership
- Attendance
- Booking
- Payment
- Trainer
- Receptionist
- Administrator
- AuthenticationController
- PaymentController

Relationships

- Member owns Membership.
- Membership contains Payment records.
- Member creates Booking.
- Attendance belongs to Member.
- Receptionist manages Members.
- Administrator manages the system.

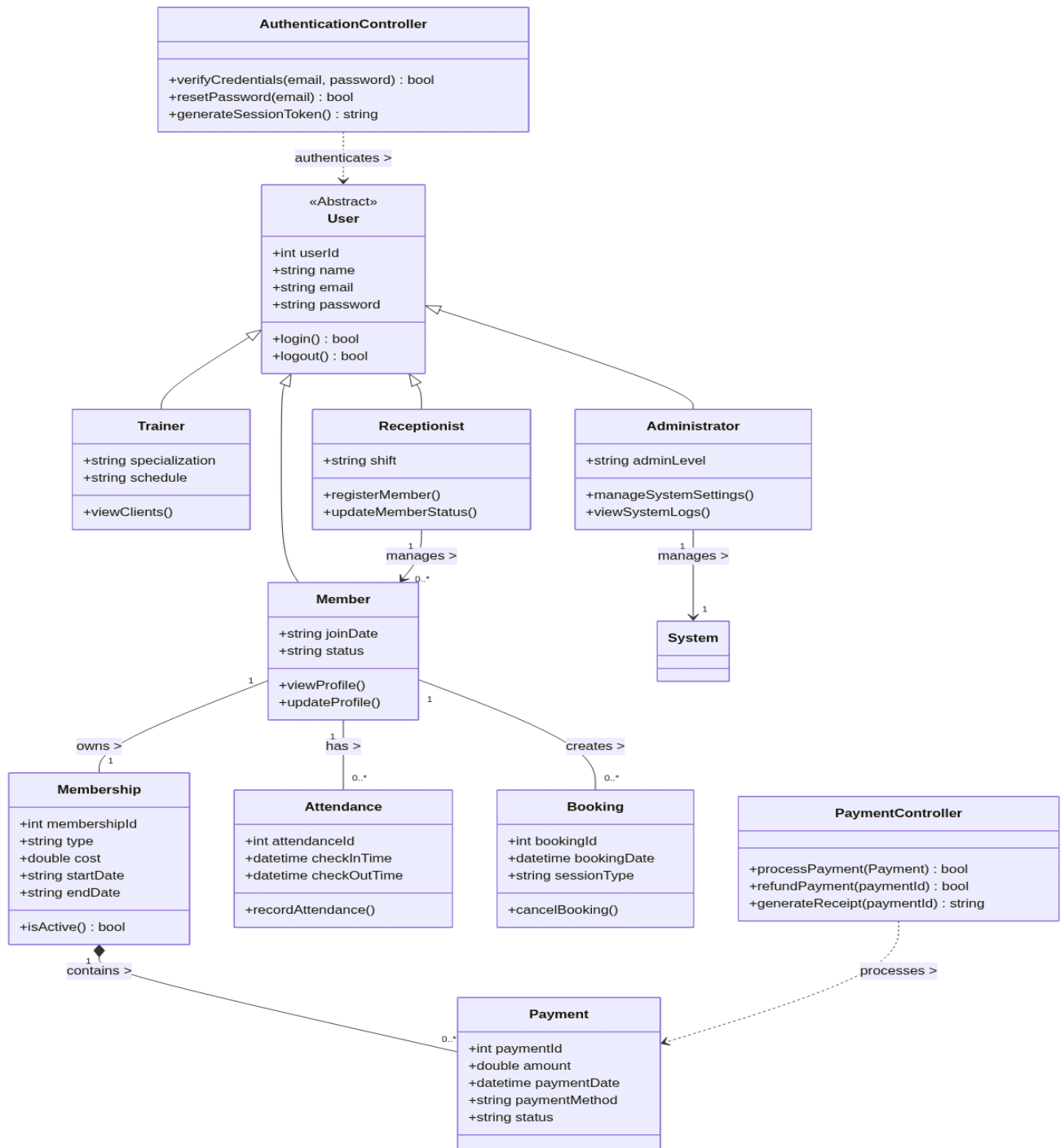


Figure 6: Class Diagram

3.6 Wireframes

The following wireframes illustrate the primary user interfaces of the application.

A wireframe of a user interface for a login and registration system. The interface is contained within a window titled "Window Title" with standard window control buttons (minimize, maximize, close) in the top right corner. The main content area is dark gray. In the center, there is a white rectangular box containing the following elements: a "User Name:" label above a text input field containing "johndoe"; a "Password:" label above a password input field containing "*****"; a blue "SIGN IN" button; a blue underlined link "Forgot Password?"; a horizontal line; the text "New User"; and a blue "SIGN UP" button.

3.7 Hardware and Software Requirements

Hardware Requirements

Client Computer

- Intel Core i3 processor or higher
- 8 GB RAM
- 250 GB storage
- Keyboard and Mouse
- Monitor

Server

- Intel Core i5 processor or higher
- 16 GB RAM
- MySQL Database Server
- Network connectivity

Optional Devices

- Barcode Scanner
- RFID Scanner
- Receipt Printer

Software Requirements

Development Software

- Java JDK 17
- MySQL 8.x
- JDBC Driver
- IntelliJ IDEA or Eclipse
- Git
- GitHub
- JUnit
- StarUML / Visual Paradigm

Operating Systems

- Windows 10
- Windows 11

4. Non-Functional Requirements

4.1 Reliability

- The Gym Management System (GMS) shall maintain an operational uptime of **99.5%** during standard facility operating hours (e.g., 5:00 AM to 11:00 PM).
- The system shall automatically execute daily database backups at 12:00 AM to prevent data loss in the event of hardware or system failure.
- In the event of a database disconnection, the local client-side terminal interface must gracefully notify the user of the connection loss without crashing.

4.2 Robustness

- The system shall validate all form inputs (e.g., membership fees, phone numbers, email structures) prior to database submission to prevent corrupt records or system crashes.
- The system must gracefully handle unexpected input errors, network timeouts, or invalid scanner inputs by displaying clear, user-friendly error messages rather than raw software stack traces.

- The authentication module must prevent brute-force attacks by locking out any user account (Admin, Trainer, or Member) for 15 minutes after 5 consecutive failed login attempts.

4.3 Performance

- Login response time shall not exceed 3 seconds.
- Member searches shall complete within 2 seconds.
- Check-in processing shall complete within 2 seconds.
- Payment processing shall complete within 5 seconds.
- Reports shall be generated within 10 seconds.

4.4 Maintainability

- The application source code shall be modular, thoroughly documented, and follow strict Object-Oriented Programming (OOP) principles to support future extensions (such as adding online personal training portals).
- All development files, database schemas, and codebase versions must be managed and synchronized securely using Git and GitHub.
- System configurations, including membership tier rates and facility details, must be easily updateable from the Admin Dashboard database settings without requiring manual code modifications.

4.5 Security

- The Gym Membership Login System shall implement strict Role-Based Access Control (RBAC), ensuring members and trainers cannot access administrative tables or financial logs.
- All user passwords must be hashed using a secure cryptographic hashing algorithm (such as bcrypt) before storage; passwords must never be saved in plaintext.
- Database transactions must utilize prepared statements and parameterized queries to mitigate SQL injection vulnerability risks.
- Any transmission of sensitive information (such as password inputs or payment processing steps) must occur over secure HTTPS connections.

4.6 Usability

- The Graphical User Interface (GUI) shall feature a simple, consistent, and intuitive design utilizing a standard color theme (e.g., clean contrast) and responsive layout frameworks.
- Navigation links and core buttons must remain uniform and predictable across all admin, trainer, and member dashboards.

- System error notifications must clearly state the issue and guide the user toward a corrective path (e.g., "*Invalid email format. Please include an '@' symbol*").
- The system interfaces shall be designed to require minimal training, allowing newly onboarded receptionists or gym staff to use the GMS with less than 30 minutes of initial guidance.

5. References

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